# Storage-Centric Systems for Genomics and Metagenomics


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SAFARI — Group's Website

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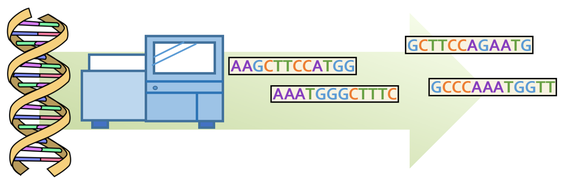
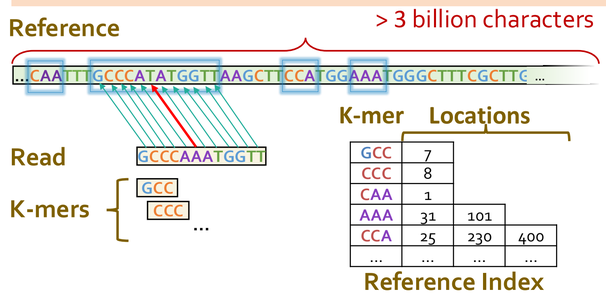
## 1: Background

### Genomics

- Genome sequence analysis is critical for many applications
  - Personalized medicine
  - Outbreak tracing
  - Evolutionary studies
- Genome sequencing machines extract smaller fragments of the original DNA sequence, known as reads




### Read Mapping Process



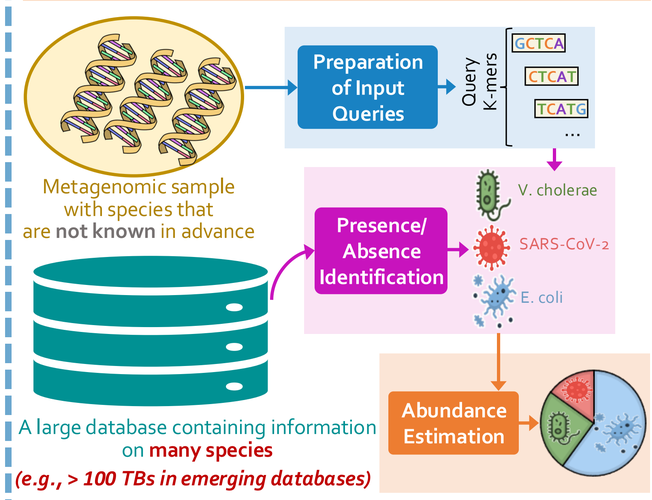

| | |
|---|---|
| Seeding | Determine potential matching locations (seeds) in the reference genome |
| Seed Filtering (e.g., Chaining) | Prune some seeds in the reference genome |
| Alignment | Determine the exact differences between the read and the reference genome |

### Metagenomics

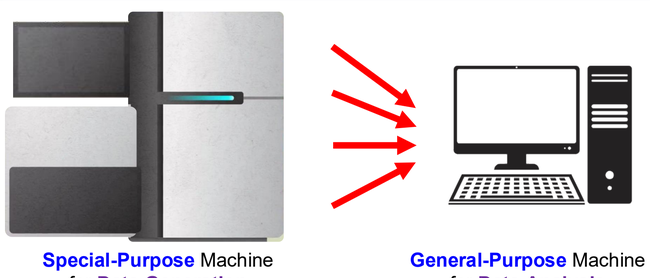
*Metagenomics:* Study of genome sequences of diverse organisms within a shared environment (e.g., blood, ocean, soil)

Has led to ground-breaking advancements

### Key Steps of Metagenomic Analysis

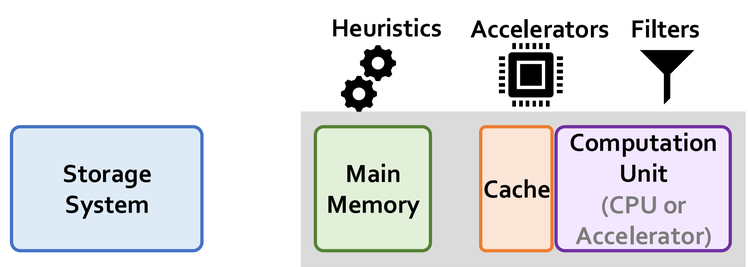



## 2: Problem with (Meta)Genomics




- ✗ Computation overhead
- ✗ Data movement overhead

## 3: Accelerating (Meta)Genomics




- ✓ Computation overhead
- ✗ Data movement overhead

## 3: Our Goal

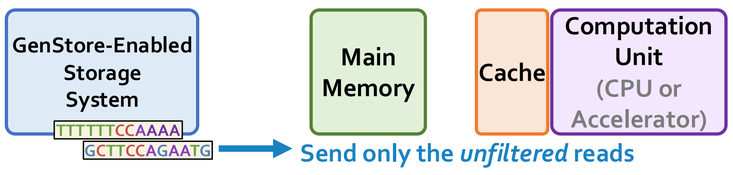
*Improve (meta)genomic analysis performance by reducing large data movement overhead from the storage system in a cost-effective manner and with high accuracy*

Challenges:

- Workloads can exhibit different behavior based on the input read set
- There are limited hardware resources in the storage system

## 3: GenStore – Storage-Centric Genomics

*Filter reads that do not require alignment inside the storage system*



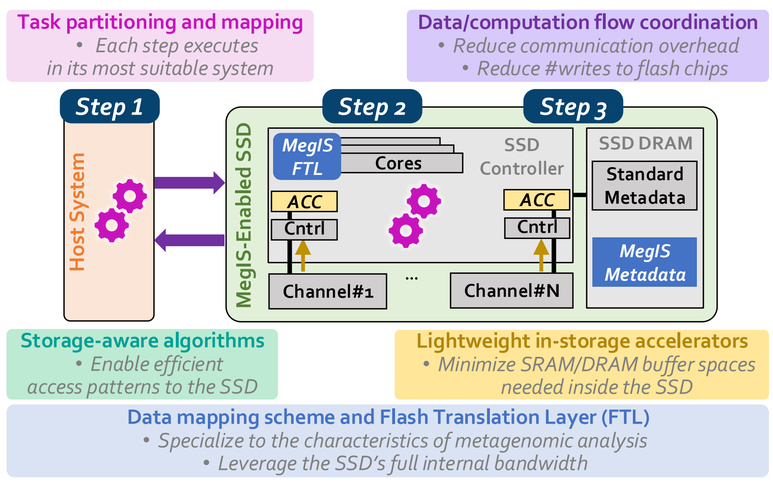

- ✓ Computation overhead
- ✓ Data movement overhead

GenStore provides significant speedup (1.4x - 33.6x) and energy reduction (3.9x – 29.2x) at low cost

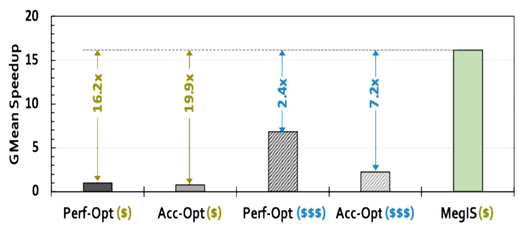
More in the paper: (ASPLOS 2022)

## 4: MegIS – Storage-Centric Metagenomics

- First in-storage system for *end-to-end* metagenomic analysis
- **Idea:** Cooperative in-storage processing for metagenomic analysis
  - Hardware/software co-design between the storage system and host system




- Cost-optimized system ($): With SSD-C and 64-GB DRAM
- Performance-optimized system ($$$): With SSD-P and 1-TB DRAM



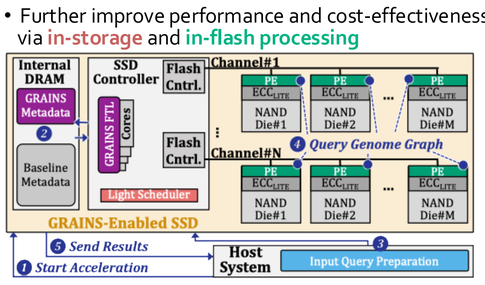

MegIS outperforms the baselines *even* when running on a much less costly system

More in the paper: (ISCA 2024)

## 5: Storage-Centric Systems for Emerging Domains

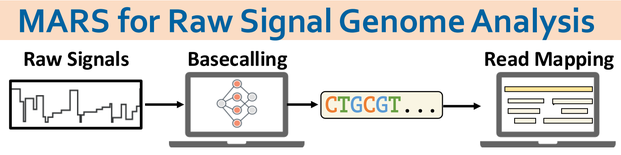
### GRAINS for Genome Graphas

- More storage-friendly analysis via specialized execution flow and scheduling
- Further improve performance and cost-effectiveness via in-storage and in-flash processing




GRAINS provides significant speedup (1.5x – 47.8x) and energy reduction (3.1x-31.6x) at low cost

### MARS for Raw Signal Genome Analysis




- ✓ Raw signals contain richer information than bases
- ✓ Efficient analysis with better scalability and portability

- MARS: First In-Storage-Processing system for *RSGA*
  - *Targeted software modifications to optimize RSGA for in-storage executions*
  - *Processing-In-Memory within the SSD to accelerate RSGA steps*

MARS provides significant speedup (28x – 93x) and significant energy reduction at low cost

More in the paper: (ICS 2025)

## 6: SAGe – Mitigating Data Preparation Bottleneck

Genomic Data is Growing at a Fast Pace

- Accelerate genome analysis
- Store genomic data in highly-compressed formats

*Data Preparation Bottleneck*

Where genomic sequence data is stored in compressed form and needs to be decompressed and formatted first before an accelerator can operate on it

An algorithm-architecture co-design for **highly-compressed storage** and **high-performance access** of large-scale genomic sequence data

- Carefully exploits properties of genomic data in synergistically co-designing algorithms and hardware

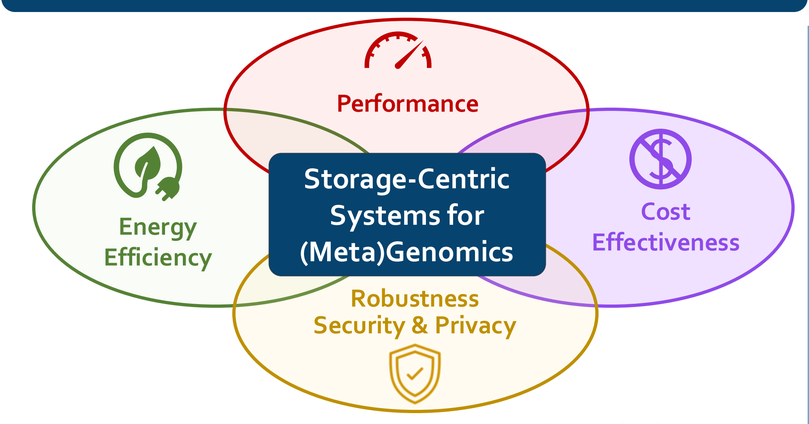
✓ High compression ratios
*Comparable to state-of-the-art genomic-specific compressors*

✓ Fast decompression using only lightweight hardware and efficient streaming accesses
*Can integrate with genomics systems even in resource-constrained environments*
- *Portable systems*
- *Near-data processing systems (e.g., in storage or memory)*

Improves the average end-to-end performance (3.0×–32.1×) and energy efficiency (13.0×–34.0×) of genome sequence analysis accelerators

More in the paper: (to appear in HPCA 2026)

## 7: Conclusion




*Make accurate (meta)genomic analysis more accessible for wider adoption*

## 8: Architecture for Health (Arch4Health) Initiative

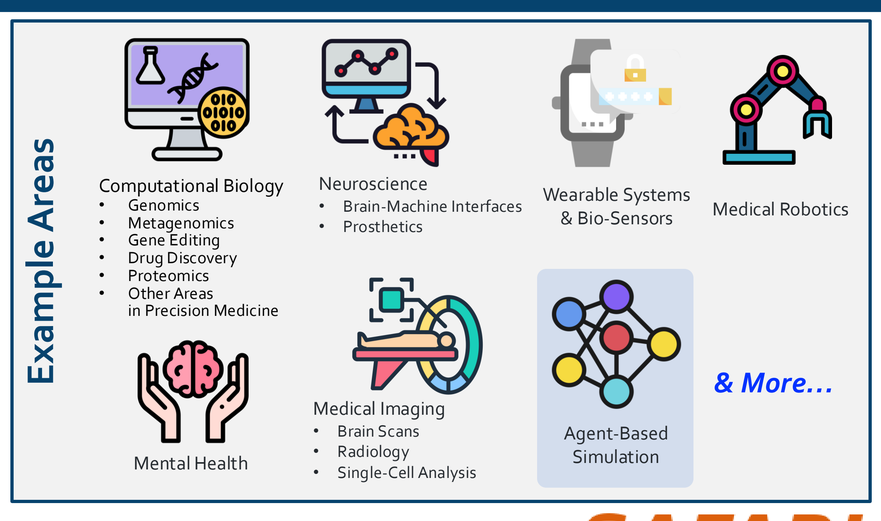
Arch4Health

*Exploring the key computational challenges in health-related applications*

&

*The vital role of computer architecture in overcoming them to advance healthcare*

Example Areas




More information:

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